

VINAYAK KOLI

+91-9711210844 | vinayak23597@iiitd.ac.in | [linkedin](#) | [GitHub](#)

EDUCATION

Indraprastha Institute of Information Technology Delhi (IIIT-D)

Bachelor of Computer Science and Social Sciences

2023- 2027

SKILLS

Programming Languages: Java, Python, C++, JavaScript, PHP , MySQL, SQL

Version Control Tools and IDE: Git, Github, Visual Studio Code , IntelliJ , PopSQL , Postman , Android Studio

Course Work: Database and Management System, Object-Oriented Programming, Advance Programming
Operating Systems, Data Structures, and Algorithms , Analytical Designing Algorithm

Work EXPERIENCE

Backend Intern | 1stop.ai

Oct 24 – Dec 24 Delhi

- **Developed a Task Manager Application** using Laravel REST APIs and Postman, enabling **CRUD operations** for tasks with user authentication achieved **100%** endpoint test coverage using Postman.
- **Delevoped a Booking Management System** with Laravel, featuring real-time booking status, role-based access control, and automated conflict detection which reduced manual booking errors and improved admin workflow efficiency.

PROJECTS

Vocal for Local (VoLo) | React Native (Expo) and Appwrite | [GitHub](#) | [Linkedin](#)

- Built a **cross-platform mobile app** with a **FastAPI-based RESTful backend**, enabling location-based discovery of nearby local stores; deployed and managed backend services on **Render**.
- Improved application performance by **~35%** through **API optimization, caching implementation, lazy loading, and render minimization**, resulting in faster screen load times and smoother navigation.
- Increased app stability by **~40%** by implementing **defensive async handling, crash protection, and fallback UI states**, significantly reducing runtime and network-related failures.

Angry Birds | Java and LibGDX | [GitHub](#) | [Linkedin](#)

- Developed a physics-based 2D game with collision mechanics and object interactions, achieving 100% accuracy in hit detection and enhancing gameplay realism levels.
- Implemented modular, object-oriented architecture to support scalability and reusability, enabling the generation of 3 unique procedural levels and boosting player replayability.
- Designed an interactive GUI with animated menus backgrounds, increasing user engagement and reducing navigation time by **~50%** during gameplay testing.

Evaluation of AI Models | Python | [GitHub](#)

- Contributing to an **ongoing research project** on **AI automation** using **Small Language Models (SLMs)** for multi-step code generation and reasoning workflows.
- Designed and implemented a **custom evaluation system** for the DeepMind Code Contests dataset, enabling automated execution, correctness checking, failure detection, and robust result logging at scale.
- Benchmarked models on DeepMind Code Contests and HumanEval datasets (165+ problems), analyzing **pass@k, execution accuracy and failure**.

ACHIEVEMENTS & EXTRA-CURRICULAR ACTIVITIES

- Secured 3rd Position in DevHaven 2.0
- Collaborated with a 10+ member team to enhance web accessibility and deliver a cancer awareness workshop impacting 100+ individuals at Tihar Jail
- Caption of Table Tennis Team for Karwaan'24
- Member of Sports Council Operations Team
- Anime, Table Tennis, Sketching